

What is Differentiation?

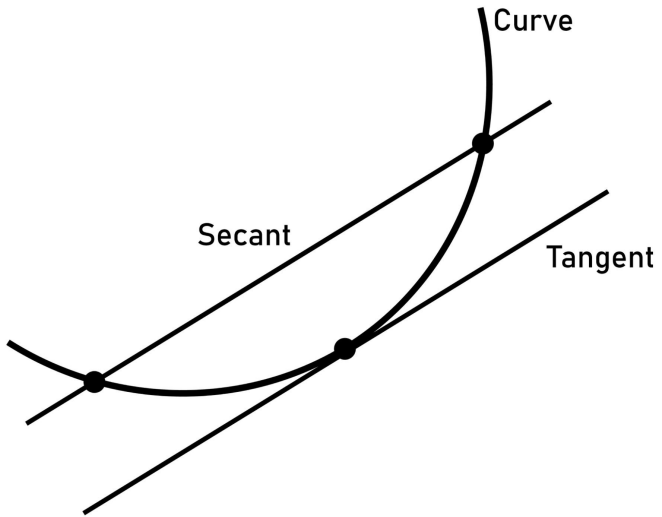
Differentiation



Average vs Instantaneous



Average vs Instantaneous





So, what is Differentiation?

Differentiation Formulas

Differentiation



Power Formula

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

$$f(x) = x^5 \implies f'(x) = 5x^4$$



Product Formula, $h(x) = f(x) \cdot g(x)$

$$h'(x) = f'(x)g(x) + f(x)g'(x)$$

$$\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$$



Chain Rule $h(x) = f(g(x))$

$$h'(x) = f'(g(x)) \cdot g'(x)$$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$



Chain Rule Example $h(x) = (3x^2+1)^5$

Inner function $g(x) = 3x^2+1$

Outer function $f(g) = g^5$

Differentiate the Inner function $g'(x) = 6x$

Differentiate the Outer function $f'(g) = 5g^4$

Multiply them:

$$h'(x) = 5g^4 \cdot 6x = 5(3x^2+1)^4 \cdot 6x = 30x \cdot (3x^2+1)^4$$



Chain Rule Example $h(x) = 0.5 * (x-a)^2$ (x-> prediction, a -> actual)

Inner function $g(x) = x - a$

Outer function $f(g) = 0.5 * g^2$

Differentiate the Inner function $g'(x) = 1$

Differentiate the Outer function $f'(g) = g$

Multiply them:

$$h'(x) = g \cdot 1 = x - a$$

Partial Derivative Formula

Differentiation



What is Partial Derivative?

$$f(x, y)$$

$$\frac{\partial f}{\partial x}$$



$$f(x, y) = x^2y + y^5$$

$$\frac{\partial f}{\partial x} = y \cdot 2x + 0 = 2xy$$

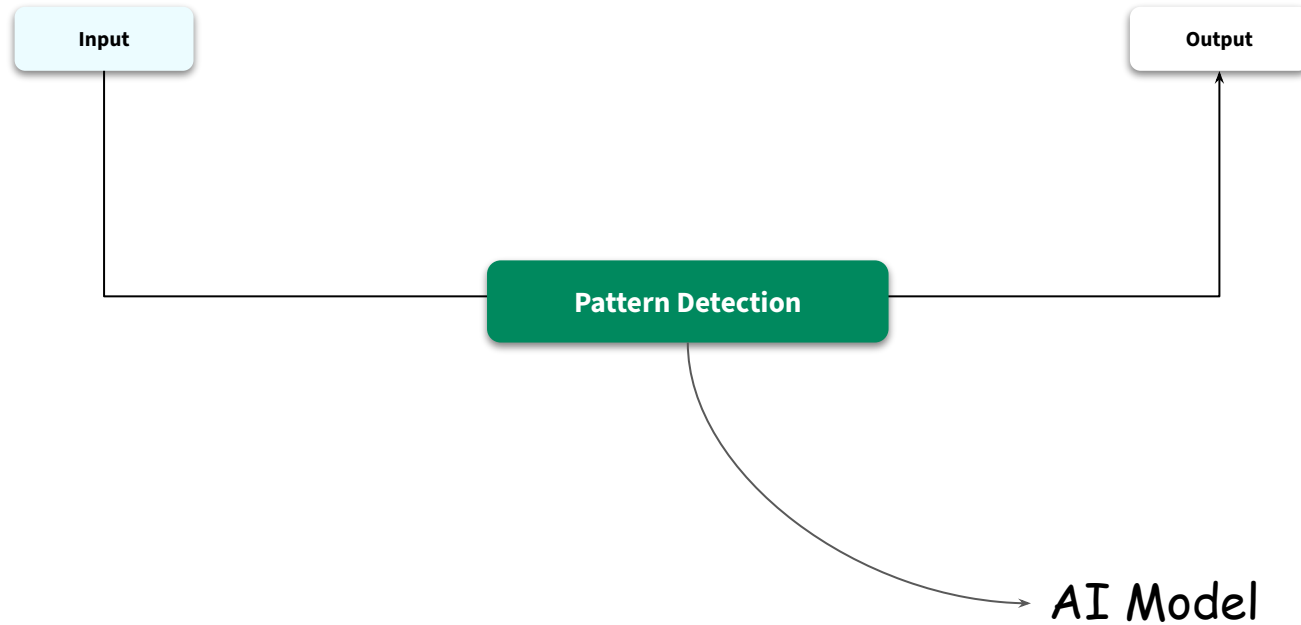
$$\frac{\partial f}{\partial y} = x^2 \cdot 1 + 5y^4 = x^2 + 5y^4$$

Differentiation in Data Science

Differentiation



How AI models are trained

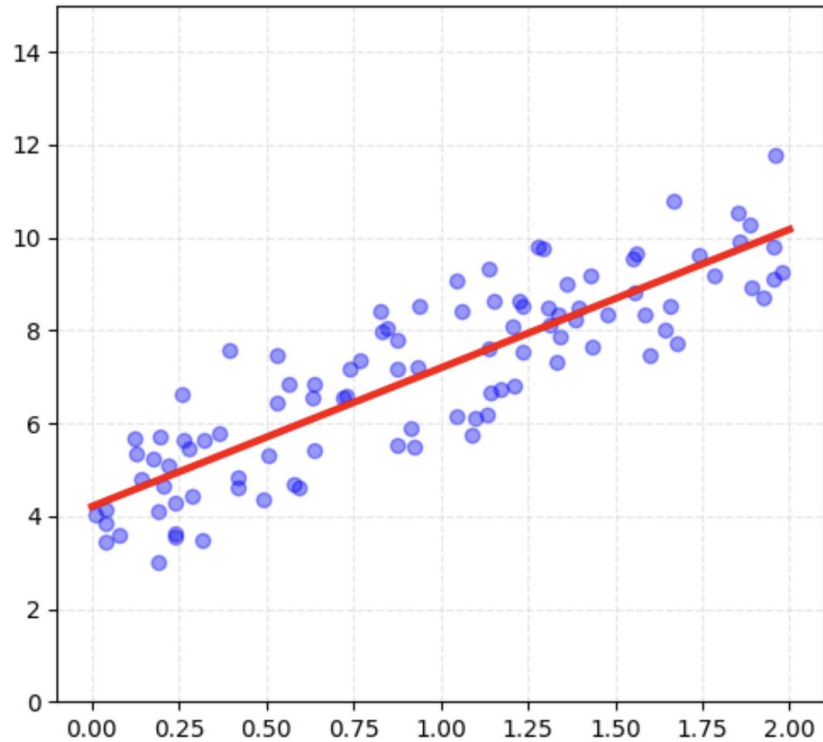




Je suis étudiant

$$\begin{array}{l}
n-1) = K = 1 \\
(\alpha \backslash x \times 2 - \frac{F+2}{2-1} \\
\cdot \sum_{i=1}^n (x = (\frac{g+1}{2} - 1) \cdot 1) \quad \frac{F}{2} = \frac{\alpha \backslash 1}{3} h \backslash k m \backslash (\pm 0) + \frac{A \backslash (14 = \pm 2 - 0)}{11} \\
l=2) = (s \backslash a) \cdot \sum_{i=1}^n s^i a^i (2 \cdot 8 \cdot (a)) \\
\backslash (\varphi \pm a \cdot a \backslash x \pm \frac{1}{2} a) \\
\frac{\overline{X}}{Y} \cdot \overline{Y} \cdot \overline{Z} \cdot (2 \cdot (1 \pm \frac{1}{2} \frac{G^2}{G_0}) \cdot \frac{1}{2} \\
\frac{a}{2} (x \cdot (a \cdot 0) + \\
\pi t \frac{1}{2} (2 \cdot a + 2 \cdot k) \cdot a \\
\leq V^2 < -b \cdot \frac{1}{0} \cdot \frac{B}{0} \cdot (F \cdot (-1)^n \cdot 1 \pm 2 \cdot c)^{\frac{1}{2} - 11} \\
\cdot f_{12} = \sqrt{4 + Y^2} \cdot g_3 \cdot 3 \cdot 4 + \frac{1 \pm 2}{2} - B \cdot F, 4 \cdot a > 16 > a \cdot (a - \\
a_1 = (K \cdot 1) = 1 \cdot x \\
\frac{1 \cdot E \cdot (e \cdot y)}{a} \\
\omega_{ab} = at_{\pm 1} \cdot e^{i \phi(a,b)} \quad \partial f(a) = (K, p) \cdot \\
h \cdot 6 \cdot (\frac{1}{2} + (4) = \frac{\sum_{i=1}^n}{\sum_{i=1}^n} + \frac{a \cdot y \cdot h \cdot f_{12}}{a \cdot y \cdot h \cdot f_{12}} \cdot f_{12}) \\
< \frac{1}{2} \cdot \frac{p}{0} \quad f = \frac{\sqrt{y \cdot K}}{0} \cdot \sum_{i=1}^n \sqrt{\frac{1}{2} - 2} \\
f) \cdot \\
a \backslash x = f \cdot c) = \frac{\sqrt{y}}{A} \cdot A = n^2 \cdot 1 \quad f(a) = 2 \cdot x \cdot (1 - (a)) \\
= = f \cdot 2(x) = \frac{K}{2} = 2 \cdot y^{\frac{1}{2}} \cdot 1 \quad R \backslash d = (x \cdot (4) + d) \\
\frac{1}{2} = -Y^2 + a^2 \leq = \frac{A \cdot c - 2}{2} \quad F(x) = (a) \cdot \frac{1}{x} - 1^2 \\
(x, p \pm) = (3) \cdot \frac{F}{4} \cdot F \cdot 4 = K \cdot (a) = (a) = \frac{0}{\pi y} \\
\partial \cdot \frac{K \cdot 1}{2} \cdot \partial a^2 + \frac{F^2}{2} \cdot \sqrt{5 \cdot g^2 \cdot (G^2 - 2 \cdot 2)} \quad F \cdot 2 = K \cdot \frac{1}{b_2} \cdot (x) + \frac{y_0}{2} = |a|
\end{array}$$

Linear Regression



$$y = mx + c$$



Linear Regression

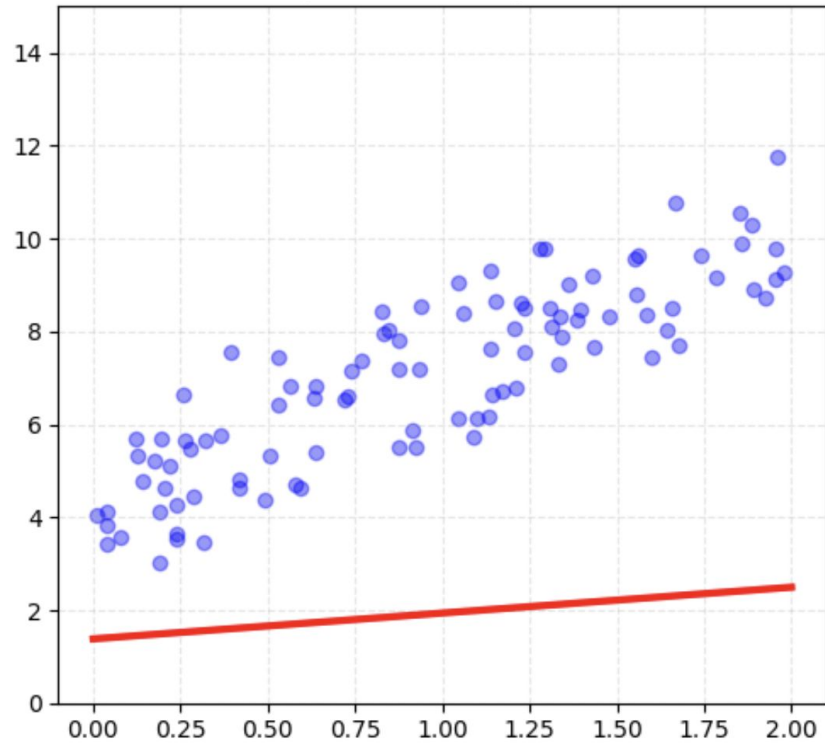
Step 1

Random Initialization

m c



Linear Regression



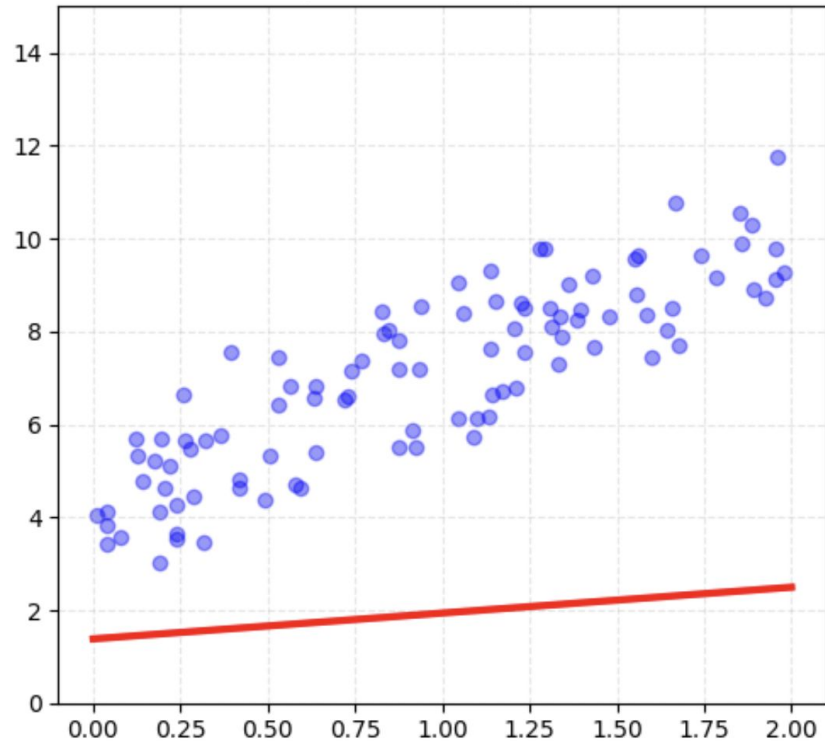
Step 2a

Calculation of $\hat{y}$

$$\hat{y} = mx + c$$



Linear Regression

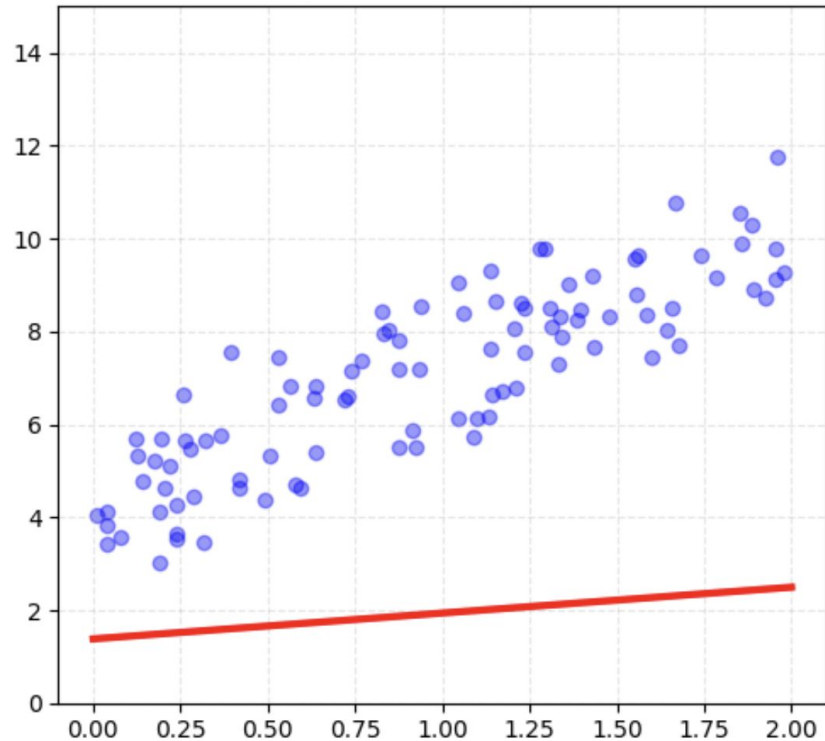


Step 2b

Calculation of Error

$$\text{Error} = (\hat{y} - y)^2$$

Linear Regression

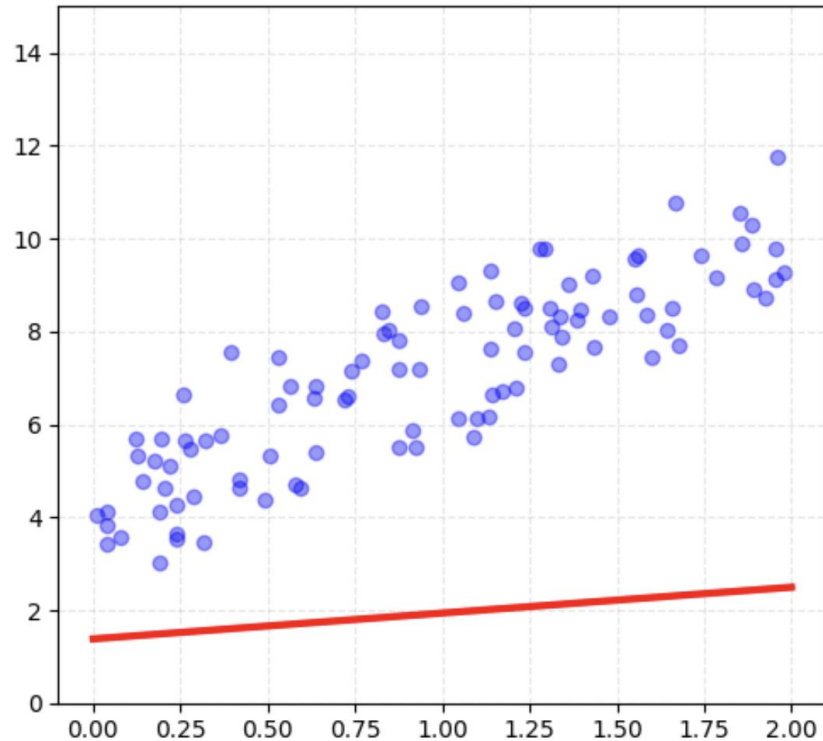


Step 2c

Calculating Gradient

$$\Delta = \frac{d(\text{Error})}{dm}$$

Linear Regression

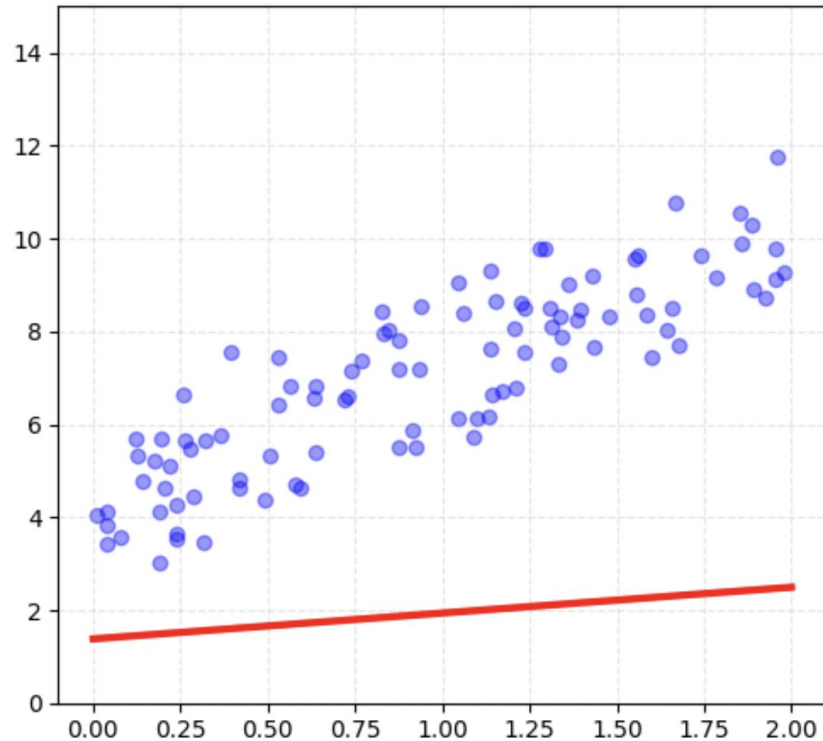


Step 2c

Calculating Gradient

$$\Delta = \frac{d}{dm} (\hat{y} - y)^2$$

Linear Regression

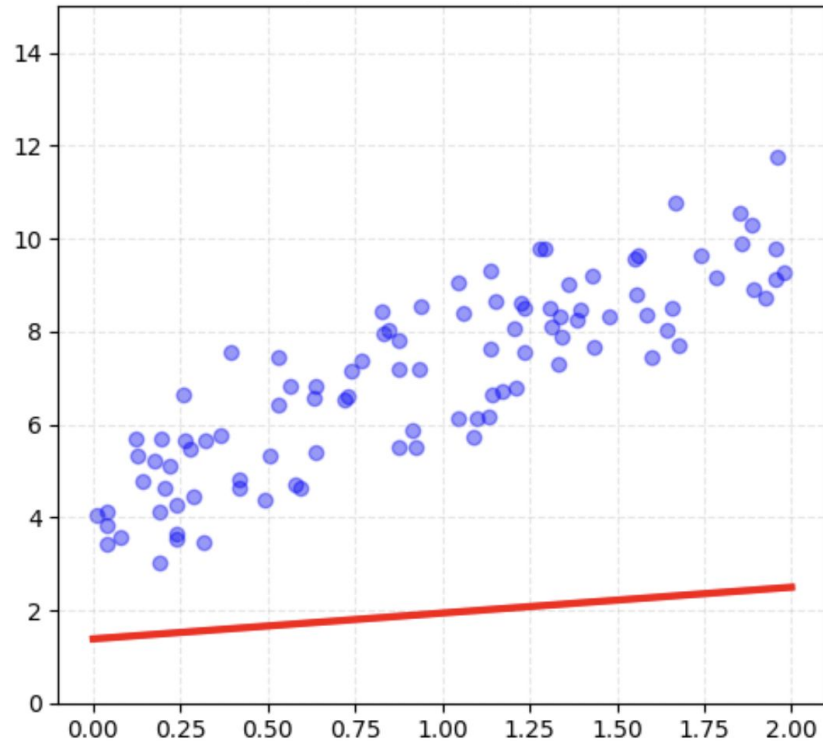


Step 2c

Calculating Gradient

$$\Delta = \frac{d (\hat{y} - y)^2}{d \hat{y}} \times \frac{d (mx + c - y)}{d m}$$

Linear Regression



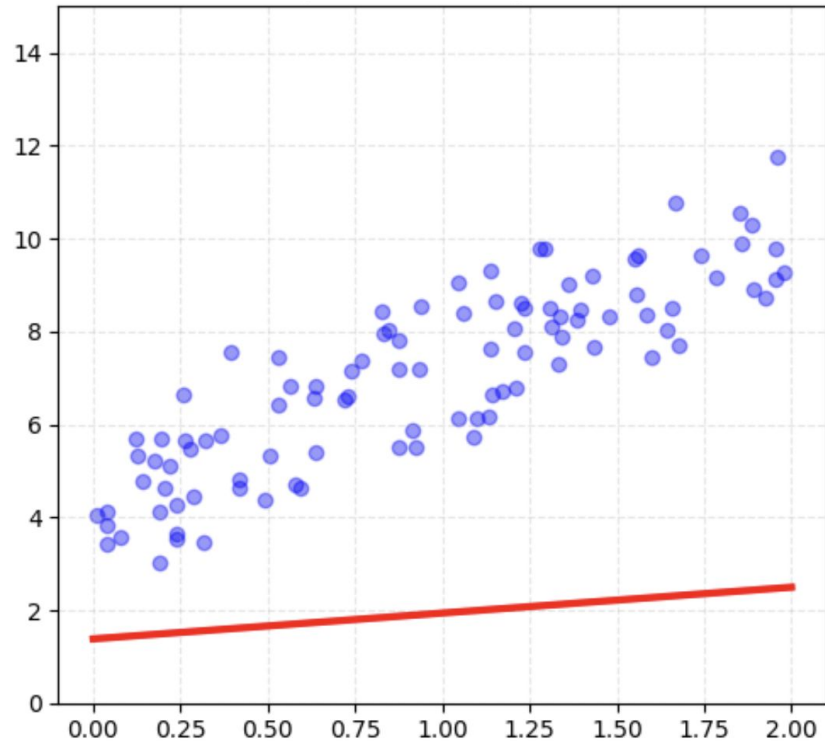
Step 2c

Calculating Gradient

$$\Delta = 2 (\hat{y} - y) \cdot x$$



Linear Regression



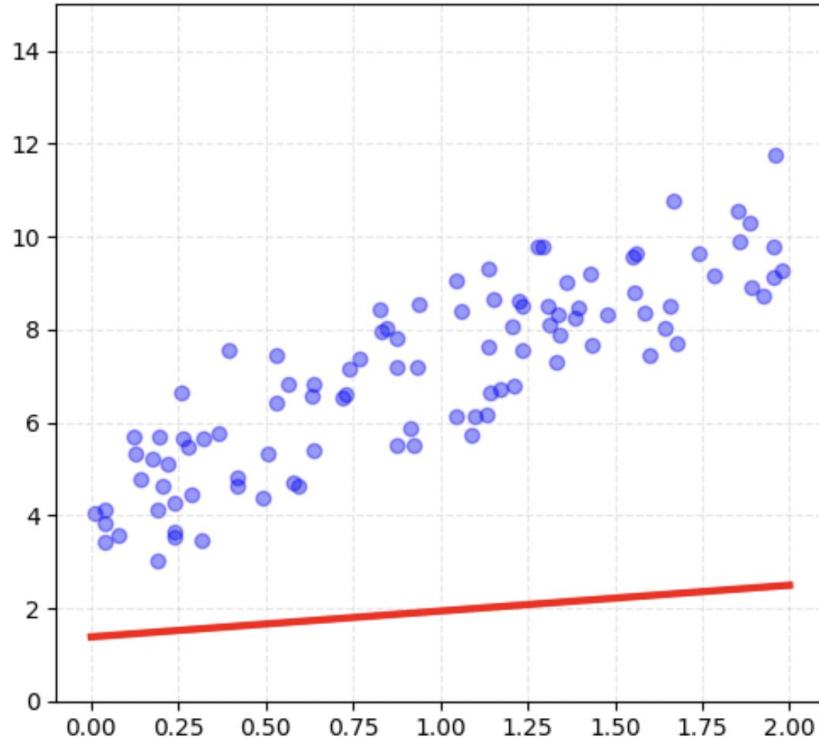
Step 2d

Updating Slope

$$m_{\text{new}} = m - a \cdot \Delta$$

$$m_{\text{new}} = m - a \cdot 2 \cdot (\hat{y} - y) \cdot x$$

Linear Regression



Step 2d

Updating Intercept

$$\Delta = \frac{d (\hat{y} - y)^2}{d \hat{y}} \times \frac{d (mx + c - y)}{d c}$$

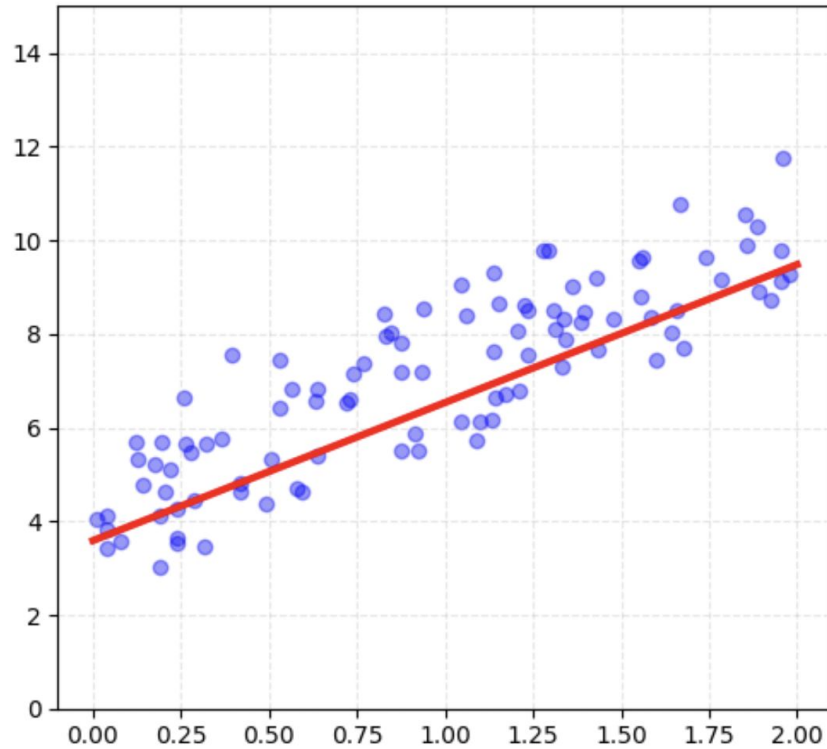
$$c_{\text{new}} = c - a \cdot \Delta$$

$$c_{\text{new}} = c - a \cdot 2 (\hat{y} - y)$$

Keep repeating loop 2a to 2d



Linear Regression



Step 3

Final Model